A GDC (graphic display calculator) is allowed. Show all working.

A reforestation project plants 500 trees in a plot at the start of 2023. Each year, 4% of the trees die, but 40 new trees are then planted at the end of the year to replace losses and expand the plot.
Let u_n be the number of trees at the start of year n, where $n = 1$ corresponds to the start of 2023, so $u_1 = 500$.

Every exercise on this test refers to this same reforestation project.

EXERCISE 1 **[9 marks]**

(a) Show that there are 520 trees at the start of 2024. **[2]**

(b) Explain why $u_{n+1} = 0.96\,u_n + 40$. **[2]**

(c) Is (u_n) is arithmetic or geometric? **[2]**

(d) Find the number of trees, correct to the nearest whole number, at the start of 2030. **[3]**

EXERCISE 2 **[8 marks]**

Let $d_n = u_{n+1} - u_n$ be the increase in the tree population during year n.

(a) Show that $d_1 = 20$. Show why (d_n) is a geometric sequence, and state its common ratio. **[4]**

(b) Write down an expression for d_n in terms of n. **[1]**

(c) Show that the tree population will approach 1000 trees. **[3]**

EXERCISE 3 [HL ONLY] **[11 marks]**

(a) Let $v_n = u_n - 1000$. Show that (v_n) is a geometric sequence. **[3]**

(b) Write down the value of v_1, and hence show that

$$u_n = 1000 - 500(0.96)^{n-1}.$$

[4]

(c) Find the first year in which the tree population exceeds 700. **[4]**

EXERCISE 4 [HL ONLY] **[11 marks]**

(a) Write down, using sigma notation, an expression for the total number of new trees planted from the start of 2023 up to and including the end of year n. **[2]**

(b) Hence find the total number of trees planted during the first 10 years. **[2]**

(c) Show that the number of trees that die during year n can be written as

$$0.04\, u_n = 40 - 20(0.96)^{n-1}.$$

[3]

(d) Hence find the total number of trees that die during the first 10 years, giving your answer correct to the nearest whole number. **[4]**

THE END